



Project Report: Concrete Strength Prediction

1. Project Overview

The goal of this project is to **predict the compressive strength of concrete** based on its ingredient composition using machine learning. We explore the dataset, preprocess it, train multiple models, evaluate them, and explain predictions using SHAP explainability tools.

2. Dataset Overview

- 📁 **Dataset Source:** Concrete compressive strength dataset
- 📏 **Samples:** 1030 observations
- 📐 **Features:**
 - Cement
 - Blast Furnace Slag
 - Fly Ash
 - Water
 - Superplasticizer
 - Coarse Aggregate
 - Fine Aggregate
 - Age
 - **Target:** strength (compressive strength in MPa)

3. Data Preprocessing

- ✓ Checked for missing values → none found
 - ✓ Removed duplicates if present
 - ✓ Capped outliers using IQR method
 - ✓ Applied log transformation to skewed features (Blast Furnace Slag, Fly Ash, Superplasticizer, Age)
 - ✓ Scaled features using StandardScaler
 - ✓ Split into training/testing sets (80/20 split)
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4. Model Selection

We used:

- Linear Regression (baseline)
 - Random Forest Regressor
 - XGBoost Regressor (best performance model)
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5. Model Training and Evaluation

We evaluated with:

- MAE
- MSE
- RMSE
- R^2 score

Model	MAE	MSE	RMSE	R^2
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Linear Regression	5.52	44.88	6.70	0.83
Random Forest	3.80	30.14	5.49	0.88
XGBoost	3.17	22.47	4.74	0.91

6. Hyperparameter Tuning

We applied GridSearchCV to XGBoost with parameters:

- `n_estimators`: 100, 200, 300
- `max_depth`: 3, 5, 7
- `learning_rate`: 0.01, 0.1, 0.2
- `subsample`: 0.8, 1
- `colsample_bytree`: 0.8, 1

✅ Best Parameters: `{'colsample_bytree': 0.8, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 300, 'subsample': 0.8}`

✅ Tuned XGBoost Performance:

- MAE: 2.96
 - MSE: 19.32
 - RMSE: 4.40
 - R^2 : 0.93
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7. Feature Importance

Based on Random Forest and XGBoost:

- Top contributors: Age, Cement, Water
 - Less important: Fly Ash, Coarse Aggregate
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8. SHAP Explainability Insights

We used SHAP to visualize both global and local model explanations.



Global SHAP Summary Plot

- **Top impact features:**
 1. Age
 2. Cement
 3. Water
 4. Blast Furnace Slag
 - Higher Age and Cement push strength predictions **up**; high Water and Slag show mixed effects.
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SHAP Bar Plot (Average Importance)

- Confirms Age, Cement, and Water as the most influential.
 - Fly Ash and Coarse Aggregate contribute the least.
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SHAP Force Plot (Individual Prediction)

- Visualizes how each feature contributed to one specific test prediction.

- Example: Age and Water increased the predicted strength, while Coarse Aggregate decreased it.

9. Comparative Analysis

Model	Strengths	Weaknesses
Linear Regression	Simple, fast, interpretable	Misses nonlinear patterns
Random Forest	Captures nonlinearities, robust	Needs tuning, less interpretable
XGBoost	Best accuracy, handles feature interactions	Needs careful tuning, more complex

 **Best performer:** Tuned XGBoost

10. Conclusion

Key Findings:

- XGBoost (after tuning) achieved best results: MAE 2.96, R² 0.93.
- Age, Cement, and Water are critical predictors.
- SHAP plots provided deep insights into how predictions are made.

Challenges:

- Handling skewed features
- Hyperparameter selection
- Visualizing complex model behavior (solved with SHAP)